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1. (a) $Cov_{AA.GM} = \rho_{A.G}\,\sigma_{AG}\,\sigma_{GG} = 0.11 \cdot 0.38 \cdot 0.18 = 0.007524$

(b) i. $\tilde{r}_p = 0.5\,\tilde{r}_A + 0.5\,\tilde{r}_G$

$\sigma_p = \sqrt{(0.5\sigma_A)^2 + (0.5\sigma_G)^2 + 2 \cdot 0.5 \cdot 0.5\,Cov_{AG}} \approx 21.9\%$

ii. $\tilde{r}_p = 0.2\,\tilde{r}_A + 0.8\,\tilde{r}_G$

$\sigma_p = \sqrt{(0.2\sigma_A)^2 + (0.8\sigma_G)^2 + 2 \cdot 0.2 \cdot 0.8\,Cov_{AG}} \approx 17.0\%$

iii. $\tilde{r}_p = 0.8\,\tilde{r}_A + 0.2\,\tilde{r}_G$

$\sigma_p = \sqrt{(0.8\sigma_A)^2 + (0.2\sigma_G)^2 + 2 \cdot 0.8 \cdot 0.2\,Cov_{AG}} \approx 31.0\%$

2. (a) $\sigma_p^2 = w^2\sigma_A^2 + (1-w)^2\sigma_B^2 + 2w(1-w)\sigma_A\sigma_B\rho_{AB}$

$= (0.4w)^2 + [0.3(1-w)]^2 + 2w(1-w) \cdot 0.4 \cdot 0.3 \cdot 0$

$= 0.25w^2 - 0.18w + 0.09$

σ_p is minimized when σ_p^2 is minimized

When $w = \frac{0.18}{0.5} = 0.36$, σ_p is minimized

$\sigma_p = \sqrt{0.25 \cdot 0.36^2 - 0.18 \cdot 0.36 + 0.09} \approx 24\%$

$\mu_p = w\mu_A + (1-w)\mu_B = 0.36 \cdot 15\% + 0.64 \cdot 7\% = 9.88\%$

(b) $\sigma_p^2 = w^2 \cdot 0.4^2 + (1-w)^2 0.3^2 + 2w(1-w) \cdot 0.4 \cdot 0.3 \cdot 0.4 = 0.154w^2 - 0.084w + 0.09$

When $w = \frac{0.084}{0.308} \approx 0.27$, σ_p is minimized.

$\sigma_p = \sqrt{0.154 \cdot 0.27^2 - 0.084 \cdot 0.27 + 0.09} \times 100\% \approx 28\%$

$\mu_p = 0.27 \cdot 15\% + 0.73 \cdot 7\% = 9.16\%$

(c) $\sigma_p^2 = w^2 \cdot 0.4^2 + (1-w)^2 \cdot 0.3^2 + 2w(1-w) \cdot 0.4 \cdot 0.3 (-0.4) = 0.346w^2 - 0.276w + 0.09$

When $w = \frac{0.276}{0.692} \approx 0.4$, σ_p is minimized

$\sigma_p = \sqrt{0.346 \cdot 0.4^2 + 0.176 \cdot 0.4 + 0.09} \times 100\% \approx 18.7\%$, $\mu_p = 0.4 \cdot 15\% + 0.6 \cdot 7\% \approx 10.2\%$

3. $u = r_f + \left(\frac{\mu_r - r_f}{\sigma_r}\right) - \left(\frac{A}{2}\right)\sigma_p^2$

$0 = \frac{\mu_r - r_f}{\sigma_r} - A\sigma_p \quad \therefore \sigma_p = \frac{\mu_r - r_f}{A\sigma_r}$ $\therefore$ optimal choice of $\sigma_p = \frac{\mu_r - r_f}{A\sigma_r}$

$\therefore$ optimal choice of $w = \frac{\sigma_p}{\sigma_r} = \frac{\mu_r - r_f}{A\sigma_r^2}$

w is inversely proportional to A, as A increases, w decreases

w is proportional to $(\mu_r - r_f)$. As $(\mu_r - r_f)$ increases, w increases.